Chapter 13. Ramsey Theory

Does It Clique?

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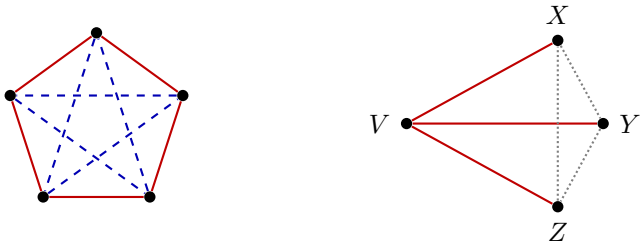
13.1 Ramsey theory for finite graphs

Motivation. Large 2-coloured cliques contain large monochromatic ones, via $R(k,\ell) \le R(k,\ell-1) + R(k-1,\ell)$.

Example 13.1

Among any six people there are three who all know each other or three who are mutual strangers.
Solution. Colour K_6 red (acquainted) / blue. A vertex V has 5 edges, so three of the same colour, say red, go to X, Y, Z. A red edge in XYZ gives a red triangle with V; otherwise XYZ is blue.

Five are not enough: pentagon sides red, diagonals blue.



K_5: no monochromatic triangle Example 13.1

Figure. *Left: K_5 without monochromatic triangles. Right: a red edge in XYZ closes a red triangle.*

Equivalently, every 6-vertex simple H or its complement contains a **clique** of size 3.

Theorem 13.2 (Ramsey's theorem for graphs)

For integers $k, \ell \ge 2$ there is a least positive integer $R(k,\ell)$ such that every red/blue colouring of the edges of $K_{R(k,\ell)}$ contains a red K_k or a blue K_ℓ.

(One such integer suffices; then a least exists.)

Example 13.3

$R(3,3) = 6$ by Example 13.1 and the pentagon. Trivially $R(2,2) = 2$.

Proof of Theorem 13.2. Induction on $k + \ell$.

Step 1 (boundary). $R(k,2) = k$: K_k is all red or has a blue K_2; likewise $R(2,\ell) = \ell$.

Step 2 (induction step). We show

$$R(k,\ell) \le R(k,\ell-1) + R(k-1,\ell). \tag{13.1}$$

In K_N, N the right side, a vertex V has $N-1$ edges: $\ge R(k,\ell-1)$ blue or $\ge R(k-1,\ell)$ red.

(a) Blue neighbours contain a red K_k or a blue $K_{\ell-1}$, which with V is a blue K_ℓ.

(b) Red neighbours contain a blue K_ℓ or a red K_{k-1}, which with V is a red K_k. ∎

Corollary 13.4

For $k, \ell \geq 2$,

$$R(k,\ell) \leq R(k,\ell-1) + R(k-1,\ell). \tag{13.2}$$

Proof. This is (13.1). ■

E.g. $R(4,3) \leq R(4,2) + R(3,3) = 10$, not sharp:

Example 13.5

$R(4,3) = 9$.
Solution.
(1) K_9 *always works.* Some V has (i) ≥ 6 red or (ii) ≥ 4 blue edges; otherwise all red degrees are 5, with odd sum 45.
(a) Six red neighbours span a triangle (Example 13.1): blue, or red giving a red K_4 with V.
(b) Four blue neighbours: all edges red give a red K_4; a blue edge gives a blue triangle with V.
(2) K_8 *need not work.* On $1,\ldots,8$ colour $\{i,j\}$ $(i<j)$ blue if $j-i \in \{1,4,7\}$, red if $j-i \in \{2,3,5,6\}$. A blue triangle with least vertex i needs two of $i+1, i+4, i+7$ (differences $3,6,3$: red). A red K_4 needs three of $i+2, i+3, i+5, i+6$, hence a pair at difference 1 (blue).

Example 13.6

$R(4,4) = 18$.
Solution. $R(4,4) \leq R(4,3) + R(3,4) = 18$. On $\mathbb{Z}_{17}$, colour $\{i,j\}$ red iff $i-j \in Q = \{\pm1, \pm2, \pm4, \pm8\}$, the nonzero squares (well defined, $-1 \in Q$).
[Supplement] *No monochromatic* K_4. Translations and multiplications by $q \in Q$ preserve the red graph, so a red K_4 normalises to $\{0,1,a,b\}$ with $a,b \in Q \cap (1+Q) \setminus \{0,1\} = \{2,9,16\}$; but differences of $2, 9, 16$ are $\pm7, \pm14 \notin Q$. Multiplication by the non-square 3 swaps the colours, so there is no blue K_4 either.

$R(4,5) = 25$, $43 \leq R(5,5) \leq 48$ (as quoted in the book); $R(k,k)$ is unknown for $k \geq 5$.

Theorem 13.7

For $k, \ell \geq 2$,

$$R(k,\ell) \leq \binom{k+\ell-2}{k-1}. \tag{13.3}$$

Proof. Induction on $k+\ell$; for $k=2$ or $\ell=2$ both sides are equal. Otherwise, by Corollary 13.4, induction and Pascal,

$$R(k,\ell) \leq \binom{k+\ell-3}{k-1} + \binom{k+\ell-3}{k-2} = \binom{k+\ell-2}{k-1}.$$

■

Corollary 13.8

$R(k,k) \leq 4^{k-1}$ for $k \geq 2$.

Proof. $R(k,k) \leq \binom{2k-2}{k-1} \leq \sum_i \binom{2k-2}{i} = 2^{2k-2}$. ■

Campos–Griffiths–Morris–Sahasrabudhe (2023): $R(k,k) < (4-\epsilon)^k$. Exponential lower bounds: Chapter 15.

13.2 Generalizations of the Ramsey theorem

Motivation. The same induction handles many colours and r-subsets; a corollary is Erdős–Szekeres.

Example 13.9

If every edge of K_{17} is coloured with one of three colours, there is a monochromatic triangle. *Solution.* V has 16 edges, six of one colour, say green, to a set g. A green edge in g closes a green triangle; otherwise g is 2-coloured on ≥ 6 vertices (Example 13.1).

Theorem 13.10

For positive integers $n_1, \dots, n_k$ there is a least $N = R(n_1, \dots, n_k)$ such that every colouring of the edges of K_n, $n \geq N$, with colours $1, \dots, k$ has, for some i, a K_{n_i} all of whose edges have colour i.

Proof (completed [Supplement]). Induction on $\sum n_i$. If some $n_i = 1$, $N = 1$. Otherwise with $R_i = R(n_1, \dots, n_i - 1, \dots, n_k)$ let

$$N = \sum_{i=1}^{k} (R_i - 1) + 2.$$

A vertex V has $\sum_i (R_i - 1) + 1$ edges, so some colour i occurs on $\geq R_i$ of them, ending in S. Then K_S has a K_{n_j} of colour $j \neq i$, or a $K_{n_i - 1}$ of colour i which with V is a K_{n_i}. ■

Colouring r-subsets instead of edges ($r = 2$):

Theorem 13.11 (Hypergraph Ramsey theorem)

Colour each r-element subset of an n-set with one of k colours, and let $n_1, \dots, n_k$ be positive integers. There is a least $N = R_r(n_1, \dots, n_k)$ such that for $n \geq N$ there are an index i and an n_i-set all of whose r-subsets have colour i.

Proof [Supplement]. Induction on r, then on $\sum n_i$.

Step 1 ($r = 1$). $N = \sum_i (n_i - 1) + 1$ by pigeon-hole.

Step 2 (trivial cases). If some $n_i < r$, any n_i-set qualifies vacuously: $N = n_i$.

Step 3 (induction step). Let $r \geq 2$, all $n_i \geq r$, $R_i = R_r(n_1, \dots, n_i - 1, \dots, n_k)$ and

$$N = 1 + R_{r-1}(R_1, \dots, R_k).$$

Fix $V \in X$, $|X| = N$, and colour each $(r-1)$-set $S \subseteq X - V$ by the colour of $S \cup \{V\}$. Some $T \subseteq X - V$, $|T| = R_i$, has all $S \cup \{V\}$ ($S \subseteq T$) of colour i. On T, either some n_j-set ($j \neq i$) is monochromatic of colour j, or some $(n_i - 1)$-set U has all r-subsets of colour i; then so does $U \cup \{V\}$. ■

Theorem 13.12 (Erdős–Szekeres)

For every n there is a least $ES(n)$ such that among any $N \geq ES(n)$ points in the plane, no three collinear, some n form a convex n-gon.

Proof. $R_3(n,n)$ points suffice. Number the points; colour a triangle red if its vertices in increasing order turn clockwise, blue otherwise. Theorem 13.11 gives n points with all triangles of one colour.

Step 1 (reduction) [Supplement]. n points are not in convex position iff one lies inside a triangle of three others: a non-hull point P lies in a triangle of a fan triangulation of the hull, not on its boundary (no three collinear).

Step 2 (no point inside a triangle). Let $\chi(x,y,z) = \pm 1$ be the orientation: cyclic-invariant, sign-changing under transpositions; a triangle $x < y < z$ has colour $\chi(x,y,z)$. If D is inside ABC,

$$\chi(A,B,D) = \chi(B,C,D) = \chi(C,A,D) = \chi(A,B,C) =: \sigma.$$

By a cyclic shift and a reflection (reversing all orientations) take $A < B < C$, all four triangles of colour σ. Then:

(a) If $D < A$: $\chi(D,A,C) = \chi(A,C,D) = -\chi(C,A,D) = -\sigma$.

(b) If $A < D < B$: $\chi(A,D,B) = -\chi(A,B,D) = -\sigma$.

(c) If $B < D < C$: $\chi(B,D,C) = -\chi(B,C,D) = -\sigma$.

(d) If $C < D$: $\chi(A,C,D) = -\sigma$.

Each gives a triangle of colour $-\sigma$, a contradiction. ■

Erdős–Szekeres: $2^{n-2}+1 \le ES(n) \le \binom{2n-4}{n-2}+1$, equality on the left conjectured (known for $n \le 6$); Suk (2017): $ES(n) \le 2^{n+o(n)}$.

13.3 Ramsey theory in geometry

Motivation. Colourings of the plane force monochromatic configurations.

Example 13.13

If every point of the plane is red or blue, some two points at unit distance have the same colour.
Solution. Two vertices of a unit equilateral triangle agree.

Example 13.14

The same holds with three colours.
Solution. Otherwise every unit equilateral triangle ABC is rainbow. Reflecting A across BC gives D with BCD unit equilateral, so D has A's colour and $|AD| = \sqrt{3}$. Hence the circle k of radius $\sqrt{3}$ about a red point is red; but k contains two points at distance 1, a contradiction.

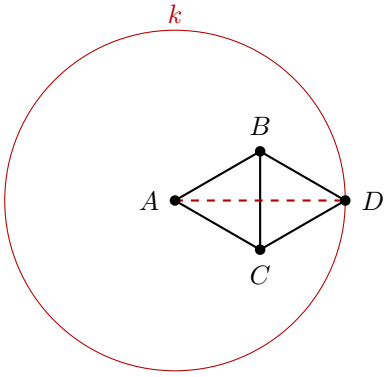


Figure. *Example 13.14: D takes A's colour, hence so does all of k.*

Example 13.15

Colour the plane red/blue, and let T be a triangle with angles $30°, 60°, 90°$ and hypotenuse 1. Some triangle congruent to T has monochromatic vertices.
Solution. Take a red unit segment AB (Example 13.13) and $D_1, \dots, D_4$ dividing the circle on diameter AB, with A, B, into six equal arcs; each $ABD_i \cong T$ (Thales). If all D_i are blue, two opposite D_i and a third form a blue copy of T.

So the **chromatic number of the plane** (of its unit-distance graph) exceeds 3; it is at most 7, and ≥ 5 (de Grey, 2018).

13.4 Selected exercises

Motivation. Pigeon-hole at one vertex, then induction, in three settings.

Exercise 13.2 (Monotone subsequences)

Prove that every permutation of length $nm+1$ has an increasing subsequence of length $n+1$ or a decreasing subsequence of length $m+1$. (The entries of a subsequence need not be in consecutive positions.)

Solution.

Step 1 (labels). Let $\lambda(x)$ be the length of the longest increasing subsequence ending at x; assume $\lambda(x) \in [n]$ for all x.

Step 2 (pigeon-hole). $nm+1$ entries, n labels: $\geq m+1$ share a label (Theorem 1.4).

Step 3 (equal labels decrease). If x precedes y with $\lambda(x) = \lambda(y)$ and $x < y$, then $\lambda(y) \geq \lambda(x) + 1$. So equal-label entries form a decreasing subsequence of length $\geq m+1$.

Sharp: 2 1 4 3 ($n = m = 2$). ■

Exercise 13.6 (Monochromatic triangles in k colours)

Let $n_k = k!(1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{k!}) + 1$. Prove that every colouring of the edges of K_{n_k} with k colours contains a triangle with monochromatic edges.

Solution. Let $m_k = n_k - 1 = \sum_{j=0}^{k} \frac{k!}{j!}$. For $k \geq 2$,

$$m_k = \sum_{j=0}^{k-1} k \cdot \frac{(k-1)!}{j!} + \frac{k!}{k!} = k\, m_{k-1} + 1.$$

Induction on k: every k-colouring of K_{m_k+1} has a monochromatic triangle.

Step 1 (initial step). $m_1 = 2$: K_3 in one colour.

Step 2 (pigeon-hole at a vertex). A vertex v has $k\, m_{k-1} + 1$ edges, so some colour, say 1, is on $\geq m_{k-1} + 1$ of them, ending in S (Theorem 1.4).

Step 3 (inside S). An edge of colour 1 in S closes a triangle with v; otherwise S uses $k-1$ colours and induction applies.

$k = 2, 3$ give $n_2 = 6$ and $n_3 = 17$ (Example 13.9). ■

Exercise 13.16 (Schur's theorem)

Colour every positive integer with one of k colours. Prove that there is an integer $N = N(k)$ such that for every $n > N$ there are integers $a, b, c < n$ of the same colour with $a + b = c$ (with $a = b$ allowed).

Solution.

Step 1 (the constant). $N = R(3, \dots, 3)$ (k entries, Theorem 13.10); let $n > N$.

Step 2 (an edge colouring). On $0, \dots, N-1$ give $\{i, j\}$ the colour of $|i - j| \in [N-1]$.

Step 3 (a monochromatic triangle). Some $i < j < l$ has one colour on its edges: $a = j - i$, $b = l - j$, $c = l - i$ agree in colour, $a + b = c$, and $a, b, c \leq N - 1 < n$. ■